

# Beekeeping and Pollination Studies

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## Land Sciences

May, 2026



## Beekeeping and Pollination Studies (A14405)

**Short Title:** Pollinators  
**Faculty** Science and Computing  
**Department** Land Sciences  
**Cluster** Horticulture  
**Author** DFLANAGAN  
**Credits** 5

**Level:** Intermediate

### Description of Module / Aims

This module aims to equip students with the knowledge and skills associated with apiculture, commercial pollination of crops and natural pollination services.

### Programmes

|           |                                   |           |
|-----------|-----------------------------------|-----------|
| HORT-0046 | BSc in Horticulture (WD_SHORT_D)  | 3 / 6 / E |
| HORT-0046 | BSc in Horticulture (WD_SHORT_DX) | 3 / 6 / E |

### Indicative Content

- The life cycle of the honey bee, bumble bee, other pollinators and their colonies
- Apiary site selection criteria, annual management programmes.
- Explanation of beekeeping equipment, assembly, and inspection of hives
- Nutrition and food sources for pollinators
- Criteria and methods of breeding
- Pest, disease, poisons and habitats of bees and other pollinators
- Key legislation, directives and action plans to protect pollinating species

### Learning Outcomes

*On successful completion of this module, a student will be able to:*

1. Associate pollinator biology, pollinator health and potential challenges.
2. Produce a plan outlining the management of a commercial apiary and protected pollinators.
3. Demonstrate routine skills associated with managing honey bees.
4. Compare the variety of forage crops for pollinators.
5. Compare attributes when selecting and breeding pollinators.
6. Relate the Sustainable Use of Pesticide Directive, and the current National Pollinator Action Plan to improved outcomes for pollinators.

### Learning and Teaching Methods

- Lectures
- Demonstration
- Field practical

## Assessment Methods

|                           | Weighing | Outcomes Assessed |
|---------------------------|----------|-------------------|
| Final Written Examination | 50%      | 1,4,5,6           |
| Continuous Assessment     | 50%      |                   |
| <i>Assignment</i>         | 20%      | 2                 |
| <i>Practical</i>          | 30%      | 3                 |

## Assessment Criteria

**<40%:** Not able to identify 40% of pollinating insect castes and equipment, unable to describe management practices and applications, unable to assemble hive, and unable to describe suitable apiary sites.

**40%-49%:** Can identify at least 40% of pollinating insect castes and equipment, can describe management practices and application in industry, able to assemble hive, and describe suitable apiary sites.

**50%-59%:** All of above and can correctly identify at least 60% of pollinating insect castes and equipment, can describe suitable apiary sites and forage requirements.

**60%-69%:** All of the above with at least 70% of pollinating insect castes and equipment identified correctly.

**70%-100%:** All of the above and at least 80% of pollinating insect castes and equipment identified correctly.

## Learning Modes

| Learning Mode        | F/T Hours | P/T Hours |
|----------------------|-----------|-----------|
| Lecture              | 36        |           |
| Practical            | 12        |           |
| Independent Learning | 87        |           |

## Essential Material

- Hooper, T. *\_Guide to Bees and Honey\_*. UK: Northern Bee Books, 2010.

## Supplementary Material

- "National Bee Unit." <http://www.nationalbeeunit.com/>
- "National Biodiversity Data Centre." <http://www.biodiversityireland.ie/>
- McMullan, J. *\_Having Healthy Honeybees\_*. Ireland: The Federation of Irish Beekeepers' Associations, 2012.
- Wilmer, P. *\_Pollination and Floral Ecology\_*. America: Princeton University Press, 2011.

## Requested Resources

- Lecture Room: Loose Seated